

eSim Semester Long Internship - Autumn 2026



**Project Title: ECG Biosignal Acquisition & Active
Filtering Front-End**

Software Used: eSim v2.3

**Layer Count: 6-Layer Printed Circuit Board (F.Cu,
In1.Cu, In2.Cu, In3.Cu, In4.Cu, B.Cu)**

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Project Overview

The **ECG Biosignal Acquisition & Active Filtering Front-End** is a multi-stage bio-potential signal conditioning system designed to capture weak cardiac signals from human skin surface electrodes. Because raw ECG voltages exist in the microvolt to millivolt range, the circuit utilizes a five-operational-amplifier cascade (X_1 through X_5) to provide high common-mode rejection, baseline wander removal, active bandpass filtering, power-line notch filtering, and output signal scaling.

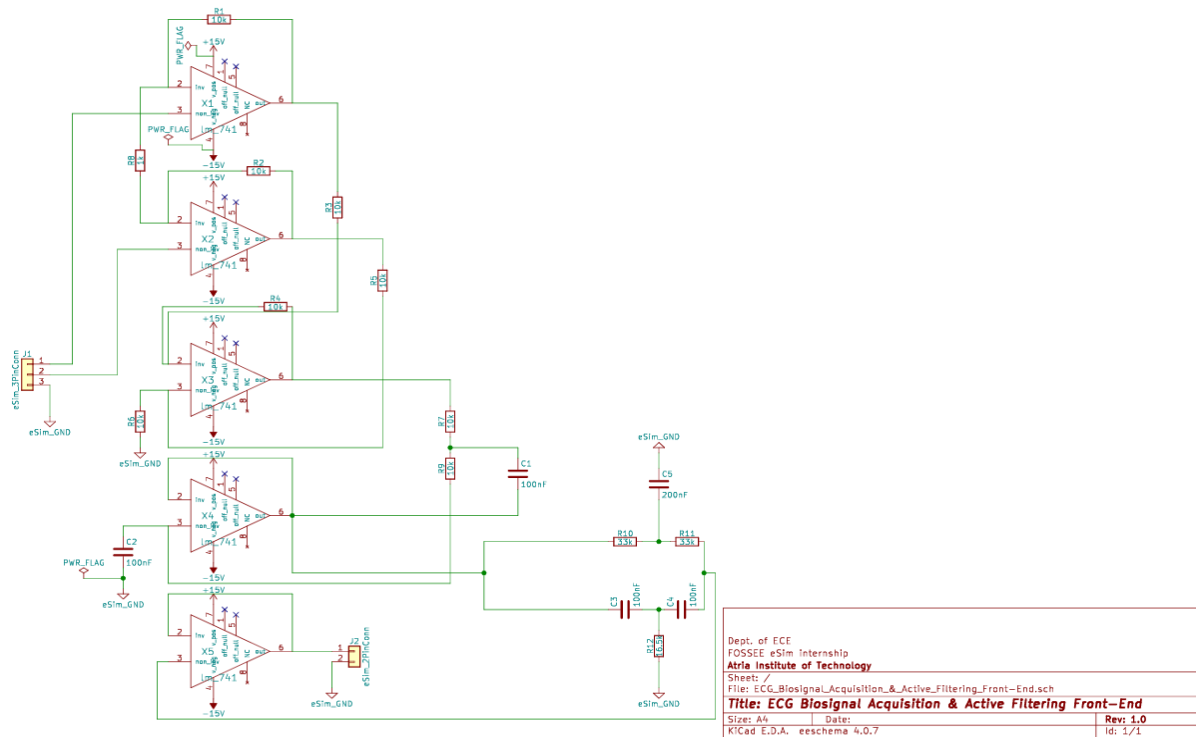
Circuit Description

The circuit consists of five primary functional stages based on the schematic architecture:

- **Power Entry Stage (J_1 & Power Flags):** Main dual-rail supply inputs (+15 V, -15 V, and GND) enter via connector J_1 . Global power distribution is routed to supply pins across all ICs with local bypass capacitors.
- **Instrumentation Differential Stage (X_1, X_2):** High-input-impedance differential buffer topology utilizing precision gain resistors (R_1, R_2, R_3, R_5, R_8). Suppresses common-mode noise while boosting differential bio-potential signals.

- **Active Bandpass Filtering Stage (X_3, X_4):** Sallen-Key active filter network configured with resistors ($R_4, R_6, R_7, R_9, R_{10}$) and capacitors (C_1, C_2, C_5). Eliminates low-frequency electrode motion artifacts (< 0.05 Hz) and high-frequency electromyographic muscle noise (> 150 Hz).
- **Power-Line Notch & Gain Stage (X_5):** Active notch filtering topology paired with output driver resistors (R_{11}, R_{12}) and capacitors (C_3, C_4) to attenuate 50 Hz/60 Hz power-line interference.
- **Output Signal Stage (J_2):** Output connector J_2 delivers the conditioned ECG analog voltage to downstream ADCs, microcontrollers, or monitoring instruments.

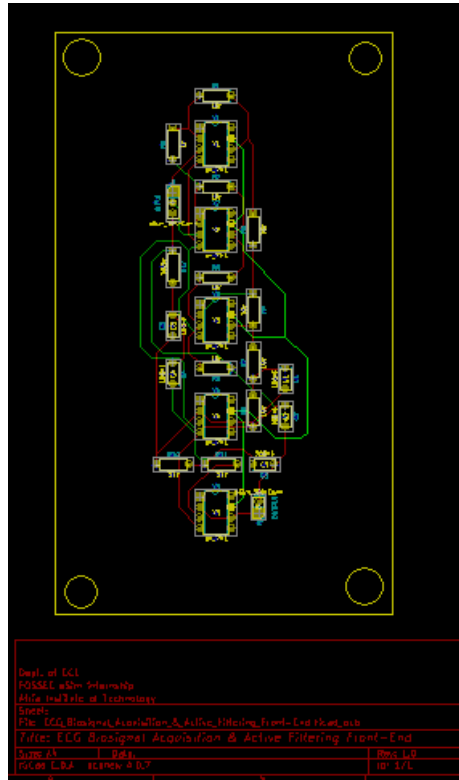
Schematic



Footprint Assignment

1	C1 -	100nF : Capacitors_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm
2	C2 -	100nF : Capacitors_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm
3	C3 -	100nF : Capacitors_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm
4	C4 -	100nF : Capacitors_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm
5	C5 -	200nF : Capacitors_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm
6	J1 -	eSim_3PinConn : Pin_Headers:Pin_Header_Straight_1x03_Pitch2.54mm
7	J2 -	eSim_2PinConn : Pin_Headers:Pin_Header_Straight_1x02_Pitch2.54mm
8	R1 -	10k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
9	R2 -	10k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
10	R3 -	10k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
11	R4 -	10k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
12	R5 -	10k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
13	R6 -	10k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
14	R7 -	10k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
15	R8 -	1k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
16	R9 -	10k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
17	R10 -	33k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
18	R11 -	33k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
19	R12 -	16.5k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
20	X1 -	1m_741 : Housings_DIP:DIP-8_W7.62mm
21	X2 -	1m_741 : Housings_DIP:DIP-8_W7.62mm
22	X3 -	1m_741 : Housings_DIP:DIP-8_W7.62mm
23	X4 -	1m_741 : Housings_DIP:DIP-8_W7.62mm
24	X5 -	1m_741 : Housings_DIP:DIP-8_W7.62mm

PCB Layout



Routing Methodology

To protect sensitive microvolt ECG signals from electromagnetic interference and power ripple, a 6-layer stackup configuration was implemented:

F.Cu(Top Layer): Dedicated to primary surface signal trace routing and component pads, kept zone-free to maintain trace isolation.

In1.Cu(Inner Layer 1): Solid Ground Plane (GND) pour providing continuous shielding and low-impedance return paths directly beneath top signal traces.

In2.Cu(Inner Layer 2): Dedicated Inner Signal Routing layer allowing trace transitions via vias to eliminate cross-overs on primary layers.

In3.Cu(Inner Layer 3): Solid Power Plane (-15V) pour supplying stable negative DC power across op-amp negative rails.

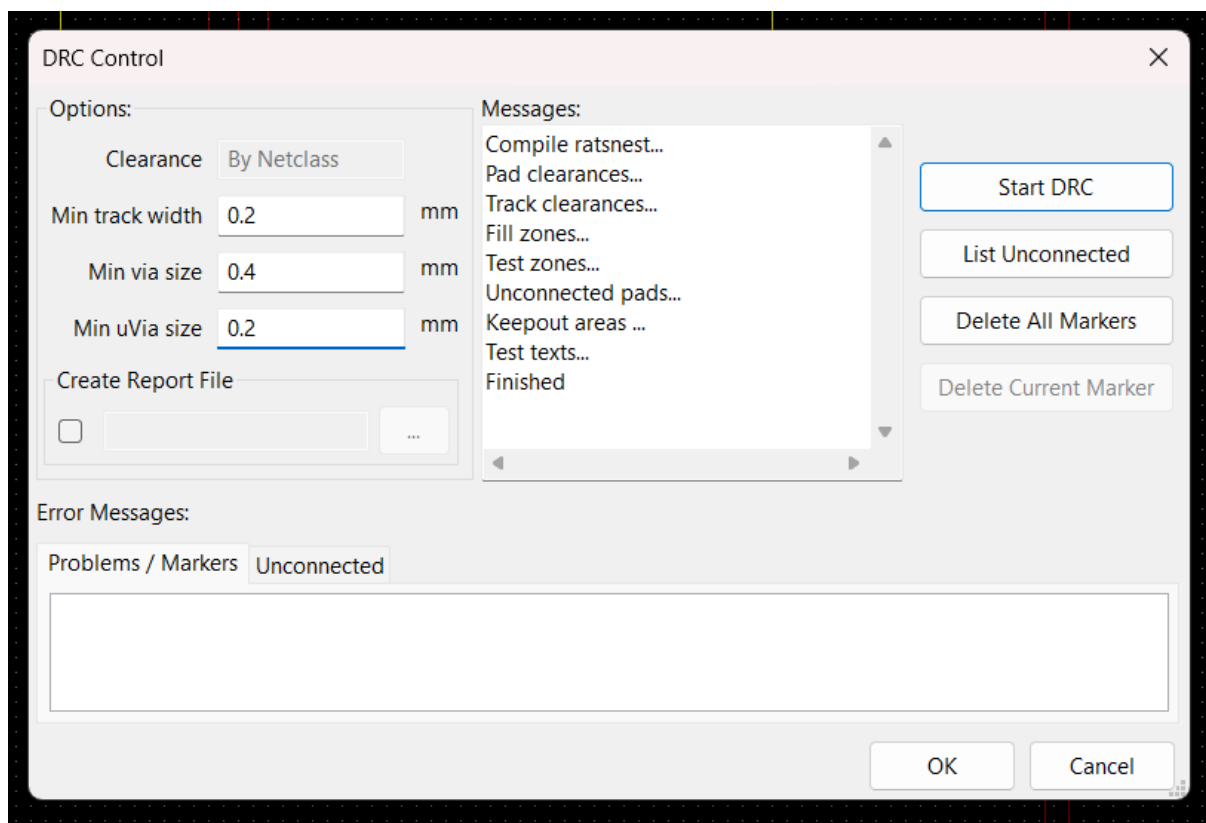
In4.Cu(Inner Layer 4): Solid Power Plane (+15V) pour supplying stable positive DC power across op-amp positive rail.

B.Cu(Bottom Layer): Auxiliary bottom signal routing for interconnecting secondary components and traces.

DRC Results

A complete Design Rule Check (DRC) was performed in PCBnew to ensure full fabrication compliance:

- Unconnected Nets: 0 (Pads: 79, Vias: 0, Track Segments: 85, Nodes: 79, Nets: 35, Links: 45, Connections: 45, Unconnected: 0)
- Clearance Violations: 0
- Short Circuits: 0



3D View

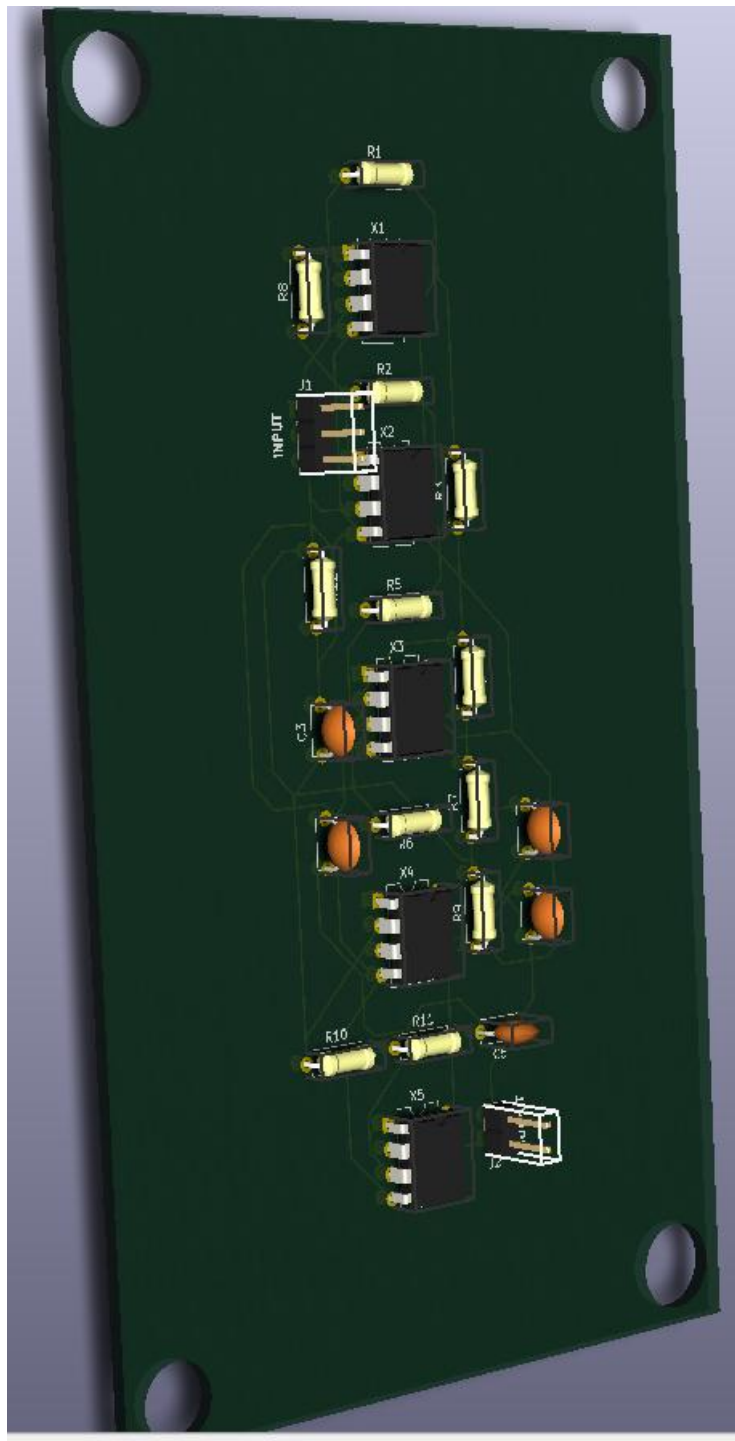


Fig: 3D isometric perspective view showing PCB substrate, board thickness, through-hole drill pads, and surface routing.

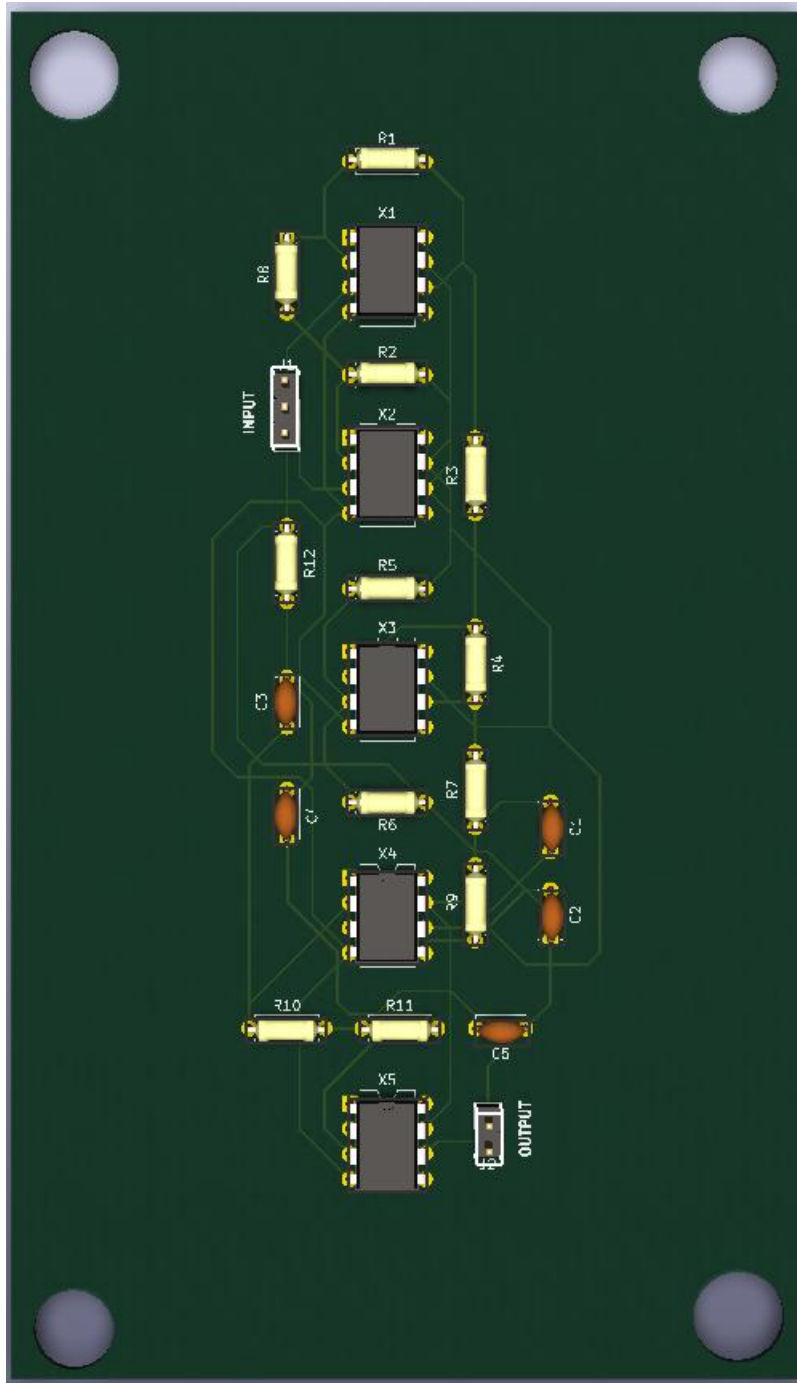


Fig: 3D top view showing silkscreen footprints, text labels, and component pad locations.

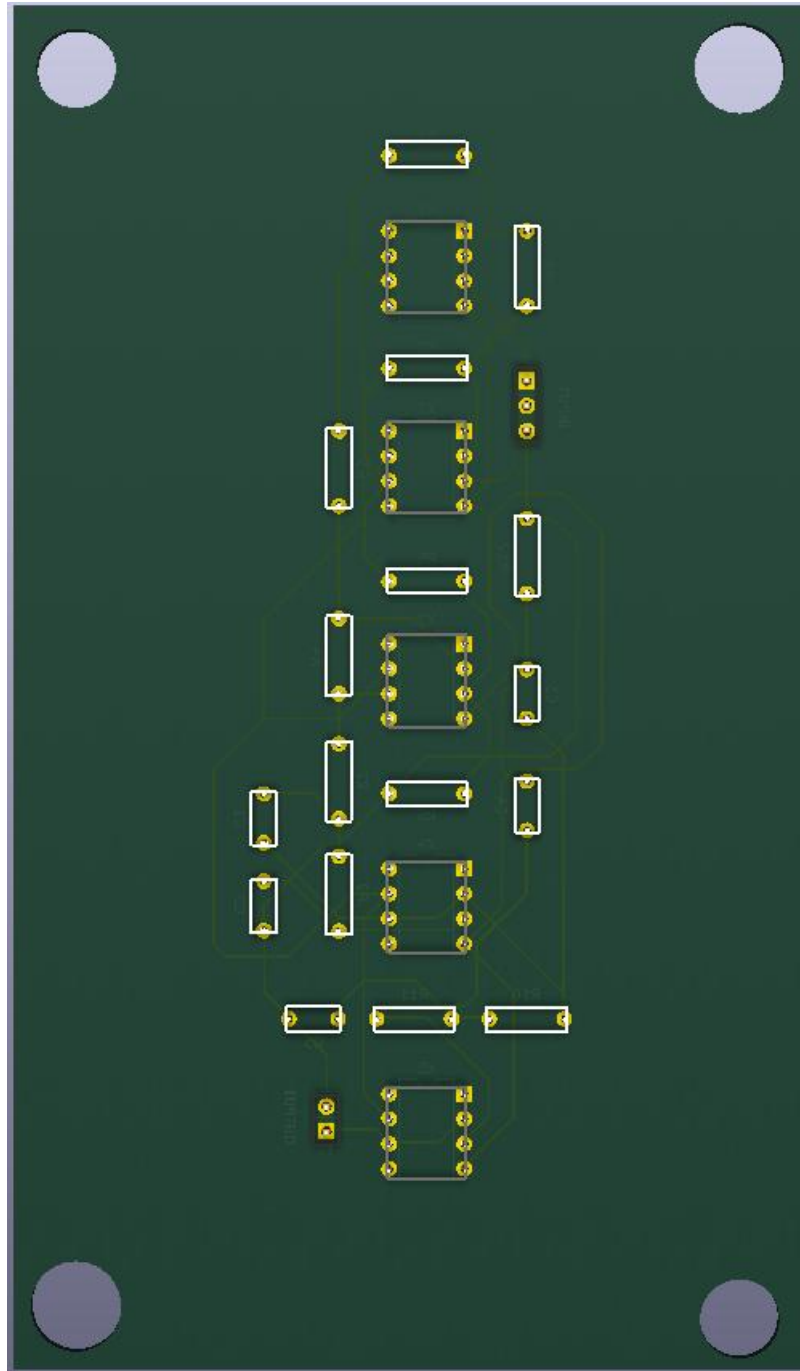


Fig: 3D bottom view showing bottom copper trace routing and terminal pads.

Design Considerations & Conclusion

- **Superior Noise Mitigation:** A dedicated **6-layer stackup** with continuous internal Ground (In1.Cu) and dual Power planes (In3.Cu -15V, In4.Cu +15V) drastically lowers loop inductance, protects sensitive microvolt-level ECG signals from EMI, and eliminates supply ripple. Decoupling capacitors ($C1 - C5$) positioned strategically along the rail paths ensure pristine signal integrity and robust high-frequency noise suppression across all five op-amp stages.
- **Prototyping & Assembly Readiness:** Footprints and silkscreen legends conform strictly to standard IPC tolerances. Highly visible reference designators ($X1 - X5$, $R1 - R12$, $C1 - C5$, $J1 - J2$), four M3 corner mounting holes, and explicit functional callouts (INPUT, OUTPUT) on the front silkscreen layer (F.Silks) enable error-free manual soldering, mechanical integration, and automated assembly.
- **Manufacturing Compliance:** Design Rule Checking (DRC) verifies zero clearance errors and zero unconnected items. Trace widths, via sizes, clearance boundaries, and 6-layer board outline dimensions (Edge.Cuts) meet all standard fabrication capabilities, making the layout fully validated for Gerber file export.

The **Multi-Stage Op-Amp ECG Signal Processing PCB** design is fully optimized and production-ready. Integrating multi-layer dual-rail power distribution, dedicated inner signal isolation layers, and a verified 6-layer stackup ensures 100%

compliance with FOSSEE/eSim standards and physical fabrication readiness.